

Homework 2

1/8/26 - 1/10/26

Skipping Q's 1 & 2 since I'm focusing on chapter 2 and completed practice problems for Ch 1.

3) Family has 3 children A, B, C

a) Is the event that A is older than B independent of the event that A is older than C?

NO, knowing that A is older than C gives new information about A's relative age. For example, now we know A is NOT the youngest sibling

b) $P(A \text{ older than } B \mid A \text{ older than } C)$

$$= \frac{P(A \text{ older than } B \cap A \text{ older than } C)}{P(A \text{ older than } C)} = \frac{2}{3}$$

4) 2 coins in a hat

└ 1 fair

└ 1 unfair $\frac{1}{4}$ H / $\frac{3}{4}$ T

Choose a coin randomly from the hat and call it coin C.

a) toss coin C twice. It shows H's both times. What is the probability C is the fair coin?

$P(C \text{ is the fair coin} \mid C \text{ lands on H twice})$

$$= \frac{P(C \text{ is the fair coin} \cap C \text{ lands on H twice})}{P(C \text{ lands on H twice})}$$

$$= \frac{\cancel{0.5} \times (0.5)^2}{\cancel{0.5} \times (0.5)^2 + \cancel{0.5} (0.25)^2} = \frac{\frac{1}{4}}{\frac{1}{4} + \frac{1}{16}} = \frac{4}{5}$$

- b) the events 'first toss of C is heads' and 'the second toss of C is heads' are NOT independent,

Knowing the first toss is heads gives information that C is more likely to be the fair coin.

- c) Find prob that in 10 flips of C, there will be exactly 3 heads

$$\begin{aligned}\text{By LOTP: } p &= 0.5 \times \binom{10}{3} (0.5)^3 (0.5)^7 \\ &\quad + 0.5 \times \binom{10}{3} (0.25)^3 (0.75)^7 \\ &\approx 0.1837\end{aligned}$$

LOTP!

- 5) woman murdered.

Her husband is the suspect.

He abused her.

1/1000 men who beat their wives end up murdering them.

1/2 of men who murdered their wives previously abused them

20% of murdered women were killed by their husbands

If a woman was murdered and the husband is NOT guilty, then there's a 10% chance he abused her.

What is the probability the man is guilty?

G: event husband murdered wife (guilty)

A: event husband abused wife

M: event wife was murdered

$$P(A|G) = 1/2$$

$$P(G|M) = 1/5$$

$$P(G|A) = 1/1000$$

$$P(A|M, G^c) = 1/10$$

$$\begin{aligned} P(G|A, M) &= \frac{P(A|G, M) P(G|M)}{P(A|M)} \\ &= \frac{P(A|G, M) P(G|M)}{P(A|M, G) P(G|M) + P(A|M, G^c) P(G^c|M)} \\ &= \frac{0.5 \times 0.2}{0.5 \times 0.2 + 0.1 \times 0.8} = \frac{5}{9} \end{aligned}$$

6) family has 2 children

assume MOB is independent of gender

B & G equally likely

all MOB equally likely

assume each sibling's characteristics are independent

O: event older sibling is a girl

Y: event younger sibling is a girl

M₀: event older sibling is born in March

M: event there's at least one female sibling born in March

$$\begin{aligned} a) P(Y|O, M_0) &= \frac{P(O|Y, M_0) P(Y|M_0)}{P(O|M_0)} \\ &= \frac{P(O) P(Y)}{P(O)} = 0.5 \end{aligned}$$

$$\begin{aligned} b) P(O \cap Y | M) &= \frac{P(O, Y, M)}{P(M)} \\ &= \frac{(0.5)^2 (1 - (\frac{1}{12})^2)}{(1 - (\frac{23}{24})^2)} \\ &= \frac{\frac{1}{4} \cdot \frac{23}{12 \cdot 12}}{\frac{47}{24 \cdot 24}} = \frac{23}{47} \end{aligned}$$



This was NOT intuitive for me

I initially thought I could discard the

1. initially ...
March information b/c of independence...

I think the solution offers a good Bayes's explanation why the March info "increases" the probability vs. $P(\text{ONLY at least one girl}) = 1/3$

★ " By symmetry, the answer should stay the same if we replaced March by another month. It's not something special about March, then, what matters. But rather the fact that the information brings 'at least one' CLOSER to being 'a specific one'. The more detailed the information being conditioned on, the closer this becomes to specifying one of the children, and thus the answer gets to $1/2$."